



امتحانات الدور الاول - ديسمبر 2025 م

المستوي الرابع - الدفعة (11) - قسم هندسة البرمجيات

اسم المادة: المعمارية الموجهة للخدمات	التاريخ: 2025/12/17 م
استاذ المادة: أ. ادريس حسين	الزمن: 3 ساعات

أجب على جميع الاسئلة

Question 1: (20 Marks)

1. What is service oriented architecture?
2. What is the characteristics of service oriented architecture?
3. What middleware component implements SOA by virtualizing and managing service interactions?
4. What is different between Verification and Validation?
5. What is different Error, faults and failures?

Question 2: (20 Marks)

1. What is Software Testing?
2. What is different between white Box Testing and Black-Box Testing?
3. There are three types of white Box Testing Techniques please mention it?
4. List the advantage and disadvantage of Black-Box Testing?
5. What is types of system testing?
6. There are three types of Unit Testing Techniques please mention it?

Question 3: (20 Marks)

Draw a cycle around only one letter that represent the correct answer

1. Which attribute listed under the "Architectural Choke Point" for monolithic systems emphasizes the tight interdependence of components?
A. Extremely complex
B. Very tightly coupled
C. Lack of standards makes it difficult to integrate
D. Rigid architecture makes even small changes complex and costly
2. In the restaurant analogy for Service-Oriented Architecture, which service is NOT mentioned as part of the system?
A. Taking orders
B. Managing inventory
C. Processing payments
D. Cleaning tables
3. Which of the following is NOT identified in the text as a key characteristic of an Enterprise Service Bus (ESB)?
A. Streamlines development
B. Supports multiple binding strategies
C. Load balancing

D. Intelligent routing

4. Which definition of SOA is attributed to IBM in the provided material?
- A. A software architecture that starts with an interface definition and builds the entire application topology as a topology of interfaces, interface implementations and interface calls
 - B. An application framework that takes everyday business applications and breaks them down into individual business functions and processes, processes, called services
 - C. A world-wide mesh of collaborating services that are published and available for invocation on a Service Bus
 - D. A design approach that emphasizes consistency, reliability, extensibility, and scalability
5. Which of the following is Key component of a service anatomy?
- A. Value Proposition.
 - B. Service Level Agreements (SLAs).
 - C. B and b are true.
 - D. A and b are wrong.
6. is a type of software testing that is carried out to test the speed, scalability, stability and reliability of the software product or application.
- A. Performance Testing.
 - B. Unit Testing
 - C. Scalability Testing
 - D. all of the above
7. It is performed to test the system beyond the bounds mentioned in the software requirements specification (SRS).
- A. True
 - B. False
8. Integration testing is the process of testing the interface between two software modules.
- A. System testing.
 - B. Integration testing.
 - C. Unit Testing.
 - D. Acceptance testing
9. is the process of checking that a software achieves its goal without any bugs.
- A. Verification
 - B. Validation
 - C. Testing
 - D. all of the above
10. is the process of checking whether the software product is up to the mark or in other words product has high level requirements.
- A. Verification
 - B. Validation
 - C. Testing
 - D. all of the above